

Qilin Huang

Pittsburgh, PA | qilinhua@andrew.cmu.edu | fanwize.github.io

Seeking Summer 2027 Internships in Computer Vision and Machine Learning

EDUCATION

Carnegie Mellon University

Master of Science in Computer Vision

Current Coursework: Advanced Computer Vision; Visual Learning and Recognition;
Introduction to Robot Learning.

Pittsburgh, PA

Expected Dec. 2027

Southern University of Science and Technology

Bachelor of Engineering in Computer Science and Technology

GPA: 3.89/4.00 | **Rank:** 13/167

Selected Coursework: Natural Language Processing; Intelligent Robotics.

Shenzhen, China

Jun. 2026

TECHNICAL SKILLS

Programming: Python, Java, C/C++, SQL.

Libraries & Tools: PyTorch, NumPy, scikit-learn, Linux, SLURM.

RESEARCH & PROJECTS

UniPixie: Controllable 3D Physics

Research Project | GRASP Laboratory, University of Pennsylvania

CVPR 2026 Highlight | First Author

Jul. 2025 – Mar. 2026

Philadelphia, PA / Remote

- Built a **conditional flow-matching model** to generate spatially varying material properties from visual features, enabling **continuous stiffness control** through scalar conditioning.
- Designed comparative experiments and ablation studies on PixieMultiVerse, demonstrating **55.6% lower MSE in log Young's modulus** than retrained PIXIE.
- Developed unified interfaces connecting model predictions to three existing simulation backends: Material Point Method, linear blend skinning, and spring-mass systems.

Neural Module Network for 3D Visual Question Answering

Research Project | Southern University of Science and Technology

Feb. – Jun. 2025

Shenzhen, China

- Designed and implemented a neuro-symbolic 3D-VQA framework via frozen-LLM program synthesis; reached 22.3 EM@1 vs. 22.4 for 3D-VisTA while improving METEOR by 6.8 points on ScanQA.
- Developed transformer-based NMN modules for filtering, spatial relations, logic, and attribute queries; the full model improved EM@1 by 1.9 points vs. mask-only and 0.7 points vs. 1-layer variants.

HCNQA: 3D Visual Question Answering

Research Project | Southern University of Science and Technology

Jun. – Sep. 2024

Shenzhen, China

ICANN 2025 | Co-Author

- Evaluated the 5×5 supervision grid and visually audited rule-generated labels for 60 questions across 6 ScanQA scenes, supporting supervision-granularity selection and identifying annotation errors and missing object references.
- Reproduced ScanQA model training and test-set evaluation, providing baseline results for HCNQA.

Face Recognition under Occlusion

Visual Computing Course Project | National University of Singapore

May – Jul. 2024

Singapore

- Co-proposed local-feature selection for masked face recognition using only unmasked training data; implemented alignment and regional feature extraction, and compared HOG and VGG-based features.